Strong Within-Domain EEG Workload Decoding Fails to Transport Across Heterogeneous Task and Device Domains: A Prospectively Evaluated Negative Methods Study

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Abstract

EEG cognitive-workload classifiers frequently achieve high accuracy *within* a single dataset, yet within-dataset performance is not evidence that a decoder *transports* to a new device, task, or cohort. We separate these two questions with a sequential, preregistration-style design across three provenance-distinct datasets: PhysioNet EEG During Mental Arithmetic Tasks (MAT; development), STEW (an exploratory, non-comparable cross-task/cross-device sensitivity dataset), and COG-BCI (Hinss et al. 2023; a single untouched prospective test). Within each development dataset, subject-baseline mean-subtraction calibration decoded workload above chance at the subject level: macro subject-level ROC-AUC was 0.880102 within MAT (full-pipeline within-subject label-permutation $p = 0.004975$, 200 permutations) and 0.839498 within STEW ($p = 0.004975$). Mean subtraction was not proven superior to absolute features in either dataset (paired confidence intervals crossed zero). When the frozen, unit-compatible pipeline was applied across datasets, the predeclared MAT→STEW mean-subtraction *transport failed* (macro subject AUC 0.447598, below chance, not better than absolute 0.472045). A post-hoc z-scoring observation on the same transport (AUC 0.682823) was the only above-chance mode; we treated it strictly as a hypothesis to be tested, froze the entire pipeline, and evaluated it *exactly once* on untouched COG-BCI. That prospective test also failed: all three calibration modes transported below chance (z-scoring 0.435961, mean subtraction 0.392153, absolute 0.367875; $n = 29$), and the primary paired contrast (z-scoring − mean subtraction) had mean $\Delta = +0.043807$ with a 95% CI $[-0.076359, +0.175233]$ that includes zero. Under the specific task/device/reference shifts tested here, baseline-relative preprocessing supported within-dataset workload discrimination but did *not* provide supported transportability of a MAT-trained workload decoder. We report this as a rigorous negative transport/reproducibility result.

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1 Introduction

Cognitive workload modulates theta- and alpha-band EEG activity during arithmetic, working-memory, and attention tasks [1,2,4]. These signatures motivate EEG-based passive brain–computer interfaces and workload monitors, but they do not by themselves establish *generalizable* prediction. Scalp EEG features also encode stable differences between participants, reference montages, electrode layouts, amplifier hardware, task timing, and recording systems. A model can therefore score well by exploiting subject- or dataset-specific structure rather than a task-general workload signal [5, 11]. The central methodological problem we address is the gap between within-dataset decoding and cross-domain transport: leave-one-subject-out (LOSO) accuracy inside one dataset is routinely high, yet that number is not a measurement of whether the same decoder works on a different device and task [24, 28].

Subject-relative calibration is a natural candidate for closing that gap. Expressing each subject's task windows relative to that subject's own resting baseline — by subtracting the baseline mean (mean subtraction) or by standardizing to baseline mean and standard deviation (z-scoring) — is a plausible route to features that are less sensitive to per-subject and per-device offsets and scales [23]. We emphasize at the outset that baseline-relative calibration is introduced here only as a candidate calibration step, **not** as a proven mechanism and **not** as a claim about neural generators.

This manuscript asks one question: does subtracting or standardizing each subject's resting EEG baseline produce a workload-decoding signal that *transfers* across provenance-distinct EEG datasets (different devices and tasks), or only *within* each dataset? We answer it with a deliberately sequential, hypothesis-controlled design. Two datasets (MAT and STEW) are used only for development; a third dataset (COG-BCI) is held completely untouched and used for a single, checksum-locked prospective test of the one hypothesis that the development phase generated. We report every decisive outcome exactly as executed, including the failures, and constrain all claims to analyses that were actually run in the repository. We do not report coupling analyses, source reconstruction, anatomical-generator claims, or any graded cross-dataset confirmation, because those analyses are not part of the verified evidence behind this paper.

2 Materials and Methods

2.1 Sequential hypothesis-control design

The study has three stages with fixed, non-interchangeable roles (Figure 1). **Development:** MAT and STEW are used to develop and stress-test the candidate calibration method and to generate hypotheses. **Post-hoc hypothesis:** an observation made *after* viewing the MAT→STEW transport is allowed to generate a single new hypothesis, which is then frozen. **Prospective test:** COG-BCI is an untouched dataset reserved for exactly one evaluation of the frozen hypothesis. MAT and STEW count only as development evidence and may not serve as prospective validation; COG-BCI is never inspected or modeled until its frozen protocol and a provenance audit have passed.

2.2 Datasets and raw-data provenance

Dataset roles, devices, sampling rates, baselines, and subject counts are summarized in Table 1.

MAT (development). The PhysioNet EEG During Mental Arithmetic Tasks dataset [4,6,26] was audited from the official raw EDF files: 36 subjects, 72 EDF files, 500 Hz, microvolt units, a standard 10–20 montage. Files ending in `_1` are rest/background and files ending in `_2` are mental arithmetic. A corrected balanced protocol was used with equal scored-rest and scored-task window counts per subject and no overlap between calibration and scored segments.

STEW (exploratory sensitivity). The STEW dataset [7] comprises 48 subjects recorded at 128 Hz with a 14-channel Emotiv EPOC+ headset under low- versus high-workload multitasking conditions. STEW is explicitly treated as a **non-comparable** cross-task/cross-device sensitivity dataset, **not** a strict replication: it differs from MAT in task, device, montage, reference, sampling rate, and amplitude units. The official IEEE DataPort STEW source was used; per-subject calibration, scored rest, and scored task were drawn from disjoint segments (14/14/14 windows per subject).

COG-BCI (untouched prospective test). The COG-BCI database [8] provides multi-session EEG including a Multi-Attribute Task Battery (MATB) and resting-state recordings on a BrainProducts system. A source provenance audit confirmed 29 readable subjects, the eight required channels, 500 Hz sampling, and the presence of eyes-open rest and MATB-difficult conditions. The primary paradigm was predeclared as MATB (highest level = MATB difficult), chosen by construct match to MAT arithmetic **before** any outcome was seen; N-back, eyes-closed rest, other MATB levels, and sessions S2/S3 were prespecified out of scope and were never analyzed.

2.3 Frozen feature space and unit-incompatibility audit

Within-dataset analyses used a frozen no-gamma 184-feature, 8-channel family. Gamma bands were removed. Cross-dataset transport, however, faces a unit problem: MAT amplitudes are in microvolts, whereas STEW values are undocumented raw 16-bit Emotiv counts for which no verified microvolt conversion exists (archive, IEEE page, and publication were all checked). We therefore restricted all transport analyses to the **96 features proven invariant under the positive affine transform** $x \to a\,x + b$ **(**$a > 0$**)**: skewness, kurtosis, Hjorth mobility, Hjorth complexity, spectral entropy, relative bandpower in δ, θ, α, β, and the θ/α, β/α, θ/β ratios, across the 8 channels. Amplitude, variance, absolute power, Hjorth activity, and Shannon entropy were excluded because they are not scale/offset invariant. The eight harmonized channels were F3, F4, F7, F8, O1, O2, T7→T3, and T8→T4.

2.4 Sampling representation (transparent amendment)

The transport-compatible feature set is scale- and offset-invariant but is **not** sampling-rate-invariant. Because the development transport pipeline (MAT→STEW) trained the source model at 128 Hz, both the source (MAT) and the target (COG-BCI) must be represented at the same rate. An earlier draft of the COG-BCI protocol stated "500 Hz native — no resample," which conflicted with the frozen 128 Hz transport representation. This was corrected **before any COG-BCI predictive metric was computed** and purely for representational consistency: both MAT and COG-BCI are resampled from 500 Hz to 128 Hz with `scipy.signal.resample_poly` (`up=32`, `down=125`) before the frozen 96 features are extracted. Native-500-Hz COG-BCI analysis was forbidden from the primary verdict. This amendment changed only the sampling representation, not the model, features, channels, endpoint, or comparator.

2.5 Calibration modes

Three feature representations were compared throughout: (1) absolute features; (2) mean subtraction relative to the subject's resting baseline (the development-selected candidate); and (3) z-scoring relative to the subject's resting baseline (a predeclared secondary diagnostic). In every transport analysis, per-subject calibration statistics were computed from the subject's **unlabeled** resting-baseline segment only, disjoint from the scored segments.

2.6 Model, evaluation, and leakage controls

The fixed model was L2-regularized logistic regression ($C = 1.0$, `solver = liblinear`, `class_weight = balanced`, `max_iter = 5000`) [21]; no label-dependent hyperparameter search was performed. The primary unit of inference was macro subject-level ROC-AUC. Within-dataset analyses used strict LOSO: in each fold the held-out subject contributed nothing to calibration statistics, imputation, scaling, or model fitting. Transport analyses fit the imputer, scaler, and model on the **source dataset only** and used target labels solely to score, never to fit, select, preprocess, or tune. Uncertainty was quantified by subject-level bootstrap confidence intervals, and within-dataset significance by full-pipeline within-subject label-permutation nulls (200 permutations, full LOSO refit per permutation). EEG I/O used MNE-Python [10, 20].

2.7 COG-BCI untouched one-shot prospective protocol

The COG-BCI test was frozen, checksum-locked, input-hash-verified, and executed exactly once. The frozen configuration trained on MAT only (rest versus arithmetic). The target was COG-BCI `ses-S1`: the unlabeled calibration baseline was the first 30 s of `RS_Beg_EO` (eyes-open rest); scored rest was the first 30 s of `RS_End_EO` (label 0); scored workload was the first 30 s of `MATBdiff` (label 1). Windows were 4 s with 50% overlap, yielding 14 rest and 14 task windows per subject; all 29 subjects were scored with no pre-outcome exclusion. The primary endpoint was the macro subject-level ROC-AUC of z-scoring; the primary comparison was the paired subject-bootstrap ΔAUC of z-scoring minus mean subtraction. The locked classification rule was: *full success* requires z-scoring above chance and a primary CI excluding zero positively; *partial* requires z-scoring above chance with the CI including zero; *failure* otherwise (z-scoring at/below chance, or not outperforming mean subtraction). No retuning, task switching, channel expansion, feature reselection, or endpoint change was permitted after outcomes were accessed.

2.8 Run-once controls and reproducibility

The frozen run materials (configuration YAML and execution script) were SHA-256-checksummed before execution; the run-time record and the live committed files were verified byte-identical. A single run marker (`RUN_MARKER.json`) records one execution, and the script aborts if any result output already exists, preventing a second result-producing run. All 29 locked input files were SHA-256-recorded in an input-hash manifest. An independent read-only integrity audit confirmed exactly one execution, internal consistency of all reported values across JSON/CSV/Markdown, alignment of checksums, and that the failure classification follows the predeclared rule.

3 Results

3.1 Provenance and pipeline validity

Raw MAT EDFs were integrity-verified (72/72 SHA-256 matched the validated MAT provenance manifest). The STEW measurement-unit ambiguity was resolved conservatively by restricting transport to the 96 positive-affine-invariant features, frozen before any transfer result was viewed. For COG-BCI, the source provenance audit passed (29/29 subjects readable, 8 locked channels, eyes-open rest and MATB-difficult present), the frozen config/script checksums matched the run-time and committed files exactly, and the one-shot script ran cleanly to completion. The leakage-free design (source-only fitting; unlabeled-baseline calibration; disjoint calibration and scored segments) was verified.

3.2 Within-MAT workload decoding is above chance

Within MAT, mean-subtraction calibration decoded rest versus mental arithmetic above chance at the subject level: macro subject-level ROC-AUC = 0.880102 (Table 2, Figure 2). The full-pipeline within-subject label-permutation null had mean 0.500600, 95% interval $[0.441057, 0.553729]$, and empirical $p = 0.004975$ (200 permutations). Mean subtraction was **not** proven superior to absolute features (0.841553): the paired mean-subtraction − absolute 95% CI $[-0.018566, +0.093112]$ includes zero. This is a within-dataset result only.

3.3 Within-STEW exploratory sensitivity

Within STEW, mean-subtraction calibration decoded low versus high workload above chance at the subject level: macro subject-level ROC-AUC = 0.839498 (permutation null mean 0.502071, 95% $[0.458830, 0.539908]$, $p = 0.004975$). As in MAT, mean subtraction was not proven superior to absolute features (0.815795; paired CI $[-0.013180, +0.060717]$ includes zero). z-scoring was numerically highest within STEW (0.898703) but is a secondary diagnostic only; per the frozen protocol the candidate method was not switched because STEW favored z-scoring. STEW remains exploratory, cross-task, cross-device, and non-comparable — supportive sensitivity evidence *inside* STEW, not replication or confirmation.

3.4 MAT→STEW mean-subtraction transport fails

This is the central negative pivot. Under the frozen, unit-invariant transport pipeline (train on MAT only, test on STEW), the predeclared candidate (mean subtraction) **failed to transport**: macro subject AUC = 0.447598, *below chance*, and not better than absolute (0.472045; paired CI $[-0.070057, +0.022646]$ includes zero) (Table 3, Figure 3). Strong within-dataset decoding (0.84–0.88) did not carry over to a different device and task.

3.5 Post-hoc z-scoring observation (hypothesis-generating only)

Of the three modes, only z-scoring transported above chance on MAT→STEW (macro subject AUC = 0.682823). Although the z-scoring computation was predeclared and verified leakage-free, its *elevation as a method of interest* occurred only *after* viewing the transport results, so it is post-hoc. We therefore treated it strictly as **hypothesis-generating**, not as confirmatory evidence: a single new hypothesis — that under severe cross-device/task scale shift, subject-level standardization might transport better than centering — to be frozen and tested once on an untouched dataset. We do not claim z-scoring transfers.

3.6 Frozen one-shot COG-BCI prospective test fails below chance

The frozen hypothesis was tested exactly once on untouched COG-BCI. All three calibration modes transported *below chance*: z-scoring 0.435961 (subject-bootstrap 95% CI $[0.359602, 0.509338]$), mean subtraction 0.392153, and absolute 0.367875 ($n = 29$; Table 4, Figure 4). The primary paired contrast, z-scoring minus mean subtraction, had mean $\Delta = +0.043807$ with a 95% CI $[-0.076359, +0.175233]$ that includes zero; the secondary contrast (z-scoring minus absolute) was $+0.068086$, CI $[-0.060173, +0.190715]$, also including zero. Under the locked rule, z-scoring is below chance, so the above-chance precondition for success or partial success is not met, and the primary superiority CI includes zero. The classification is **failure**: the MAT→STEW z-scoring elevation did not replicate.

3.7 Final decision

Combining the predeclared MAT→STEW mean-subtraction transport failure with the prospective COG-BCI failure of the post-hoc z-scoring hypothesis, cross-device/cross-task **transfer** of the MAT-trained decoder is **not supported** under the tested domain shifts, for any of absolute, mean subtraction, or z-scoring calibration. The within-dataset MAT and STEW findings are unchanged and remain within-dataset evidence only. The project verdict was recorded as `PROSPECTIVE_ZSCORE_TRANSPORT_FAILED_WRITE_NEGATIVE_METHODS_PAPER`.

Table 1: Datasets and their fixed roles in the sequential design.

Dataset	Role	Task / contrast	Baseline	Device / sampling	Subjects
MAT	Development	Rest vs. mental arithmetic	Resting block	10–20, μV, 500 Hz	36
STEW	Exploratory sensitivity	Low vs. high workload	Low-load segment	Emotiv EPOC+ 14ch, counts, 128 Hz	48
COG-BCI	Untouched prospective test	Eyes-open rest vs. MATB-difficult	Eyes-open rest	BrainProducts, μV, 500 Hz→128 Hz	29

STEW units are undocumented raw 16-bit counts; cross-dataset transport therefore used only the 96 scale/offset-invariant features. COG-BCI was resampled to 128 Hz to match the frozen transport representation.

Table 2: Within-domain subject-level decoding (L2 logistic regression, LOSO). Mean subtraction is above chance within each dataset but not proven superior to absolute features (paired CIs include zero). z-scoring within STEW is a secondary diagnostic only.

Dataset	Calibration	Macro subject AUC	Permutation p
MAT	absolute	0.841553	—
MAT	mean subtraction	**0.880102**	0.004975
MAT	z-scoring (diagnostic)	0.816327	—
STEW	absolute	0.815795	—
STEW	mean subtraction	**0.839498**	0.004975
STEW	z-scoring (diagnostic)	0.898703	—

Paired mean-subtraction − absolute 95% CIs: MAT $[-0.018566, +0.093112]$; STEW $[-0.013180, +0.060717]$ (both include zero).

4 Discussion

The principal finding is methodological: strong within-domain EEG workload decoding did **not** imply transportability. Mean-subtraction calibration decoded workload above chance within MAT (0.880) and within STEW (0.840), with permutation $p \approx 0.005$ in each dataset, yet the predeclared MAT→STEW transport of that same calibrated pipeline fell below chance (0.448). A post-hoc z-scoring observation that looked promising on STEW (0.683) did not survive a single honest prospective test on untouched COG-BCI, where every calibration mode transported below chance and the z-scoring advantage over mean subtraction was statistically indistinguishable from zero. High LOSO accuracy inside a dataset is thus not evidence of device/task generalization.

We make no neural-mechanism claim. The features that drive within-dataset decoding (band ratios, Hjorth parameters, spectral entropy) are not interpreted physiologically, and no source-localization, anatomical-generator, or coupling analysis was performed. Task, device, reference montage, amplitude units, and cohort all differ across the three datasets; we treat these as plausible, non-exclusive contributors to the transport failure, **not** as proven causes, and the present analyses do not adjudicate among them. Below-chance transport may reflect an anti-aligned decision geometry across domains, but we do not interpret it mechanistically.

The contribution is the discipline of the test, not a positive result. We froze the pipeline before the prospective evaluation, controlled leakage by fitting only on the source dataset and calibrating from unlabeled baseline segments, restricted transport to scale/offset-invariant features given undocumented STEW units, harmonized the sampling representation transparently, and executed the prospective test exactly once under checksum locks. Honest prospective falsification of a post-hoc

Table 3: MAT→STEW exploratory transport (train MAT, test STEW; 96 scale/offset-invariant features; 48 target subjects). The predeclared candidate (mean subtraction) transports below chance; z-scoring is the only above-chance mode and is post-hoc/hypothesis-generating only.

Calibration	Macro subject AUC	Above chance?
absolute	0.472045	No
mean subtraction (candidate)	**0.447598**	No (below chance)
z-scoring (post-hoc diagnostic)	0.682823	Yes

Paired mean-subtraction − absolute 95% CI $[-0.070057, +0.022646]$ (includes zero).

Table 4: COG-BCI untouched one-shot prospective test (MAT→COG-BCI; 29 target subjects). All three modes transport below chance; the primary paired contrast includes zero. Classification: failure.

Method	Role	Macro subject AUC	Subject 95% CI	Above chance?
z-scoring	primary	0.435961	$[0.359602, 0.509338]$	No
mean subtraction	primary comparator	0.392153	$[0.310697, 0.476258]$	No
absolute	secondary comparator	0.367875	$[0.286590, 0.456725]$	No

Primary ΔAUC (z-scoring − mean subtraction): mean +0.043807, 95% CI $[-0.076359, +0.175233]$ (includes zero).
Secondary ΔAUC (z-scoring − absolute): mean +0.068086, 95% CI $[-0.060173, +0.190715]$ (includes zero).

rescue is informative for reproducible BCI methodology: it bounds what subject-baseline calibration can be expected to do across heterogeneous domains.

4.1 Limitations

First, the cross-domain shift is large and confounded: device and task differ simultaneously between MAT, STEW, and COG-BCI, so the two cannot be separated. Second, there is a single prospective target (COG-BCI), tested once by design; this maximizes inferential honesty but limits breadth. Third, the model and feature family are fixed (one linear classifier; one feature set; eight harmonized channels), so the result speaks to this pipeline, not to all possible decoders. Fourth, STEW is an exploratory, non-comparable sensitivity dataset with undocumented amplitude units, not a strict replication. Fifth, the analysis was restricted to MATB-difficult versus eyes-open rest; N-back, other MATB levels, eyes-closed rest, and COG-BCI sessions S2/S3 were prespecified out of scope and contribute no results. Sixth, no permutation test was computed for the COG-BCI run (outside the frozen script's scope); it is not needed because the primary endpoint was already below chance. Finally, the study supports no clinical, deployment, diagnostic, real-time, mechanistic, anatomical, or causal claim, and reports no estimate of publication likelihood.

5 Conclusion

Across three provenance-distinct EEG datasets, baseline-relative processing supported **within-domain** workload decoding (mean-subtraction macro subject AUC 0.880 within MAT and 0.840 within STEW, each $p \approx 0.005$), but neither mean subtraction nor the prospectively tested z-scoring

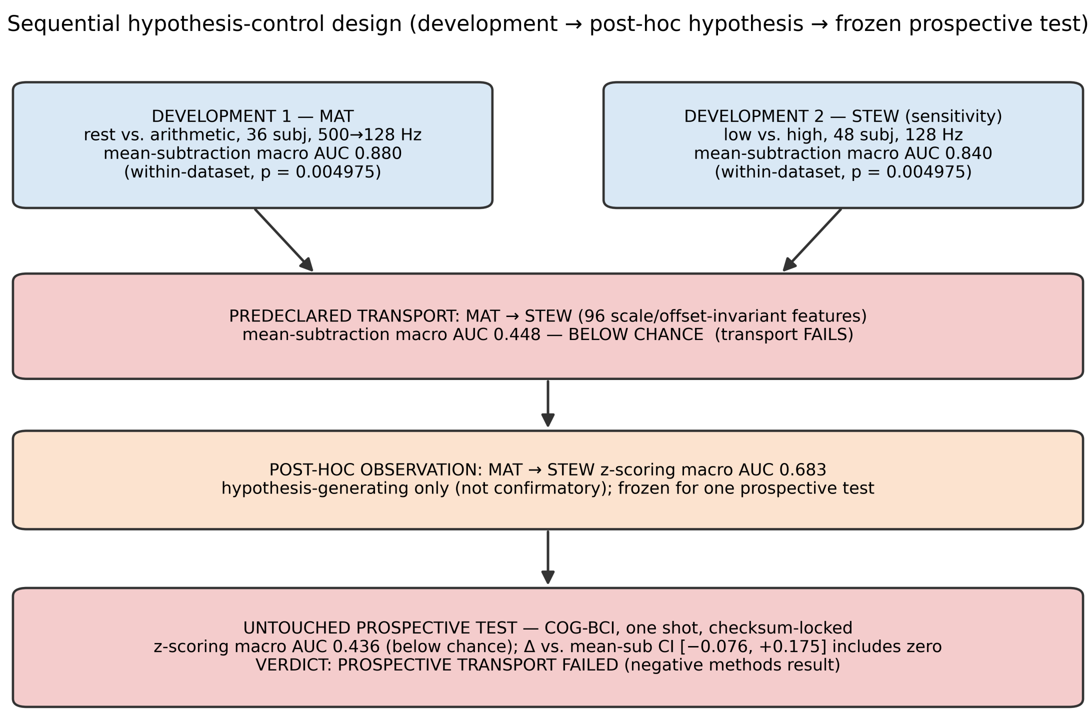

Figure 1: Sequential hypothesis-control design. Two development datasets (MAT, STEW) yield strong within-dataset decoding; the predeclared MAT→STEW mean-subtraction transport fails below chance; a post-hoc z-scoring observation generates a single hypothesis that is frozen and tested exactly once on untouched COG-BCI, where it also fails below chance.

supported **cross-domain** transport of the MAT-trained decoder. The predeclared MAT→STEW mean-subtraction transport failed below chance (0.448), and the frozen one-shot COG-BCI prospective test of the post-hoc z-scoring hypothesis also failed below chance (0.436; primary ΔAUC CI includes zero). Under the specific task and device shifts tested here, strong within-dataset decoding did not establish a transportable workload decoder. We report this as a rigorous negative transport/reproducibility result.

6 Data and Code Availability

The original datasets are available from PhysioNet EEGMAT [6], IEEE DataPort STEW [7], and the COG-BCI database [8]. Raw EEG signals are not redistributed here. The analysis code, frozen configurations, checksum manifests, and committed result-summary files are stored in the project repository; the decisive summaries are `MAT_MEAN_SUBTRACTION_SUBJECT_LEVEL_RESULTS.md`, `MAT_MACRO_SUBJECT` `STEW_EXPLORATORY_SENSITIVITY_RESULTS.md`, `MAT_TO_STEW_EXPLORATORY_TRANSFER_RESULTS.md`, `COG_BCI_ONE_SHOT_PROSPECTIVE_RESULTS.md`, and the artifacts under `results/cog_bci_one_shot/` (including `RUN_MARKER.json` and `config_and_script_checksums.json`).

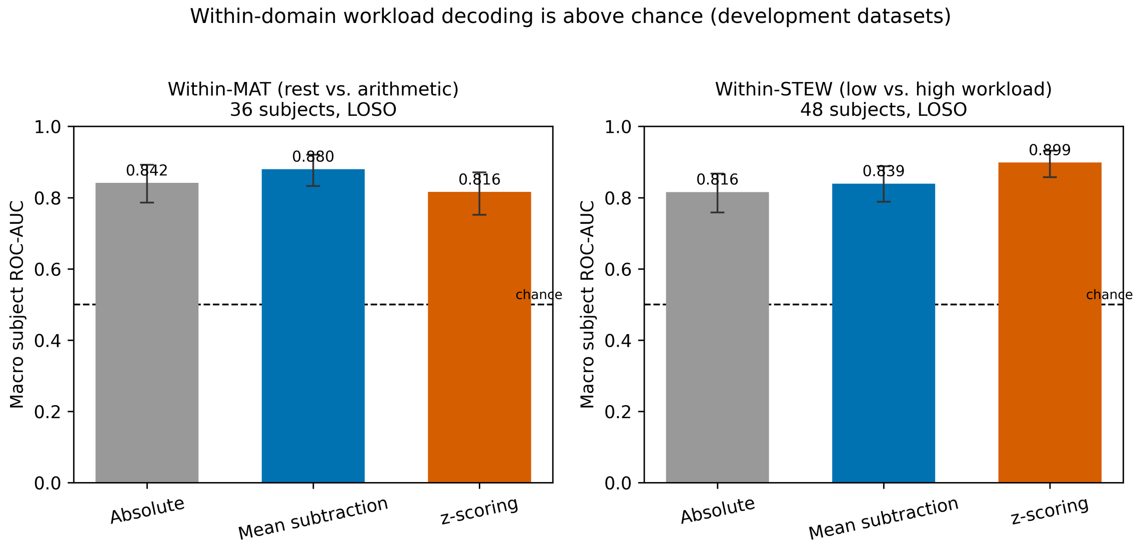

Figure 2: Within-domain macro subject-level ROC-AUC by calibration mode for MAT (left) and STEW (right). Error bars are subject-bootstrap 95% CIs; the dashed line marks chance (0.5). Mean-subtraction permutation $p = 0.004975$ in both datasets (200 permutations).

7 Ethics Statement

This work is a secondary analysis of publicly available, de-identified EEG datasets. No new human-subject data were collected. The original datasets were collected under their respective ethics approvals and consent procedures. The present work is intended for computational and methodological research, not clinical diagnosis or individual assessment.

8 Author Contributions

Arush Ravipati and Abhijay Gangarapu contributed equally to the conception, analysis, interpretation, and manuscript preparation for this study.

9 Conflict of Interest

The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest.

10 Acknowledgments

The authors thank the creators of the PhysioNet EEGMAT, IEEE DataPort STEW, and COG-BCI datasets for making their data publicly available.

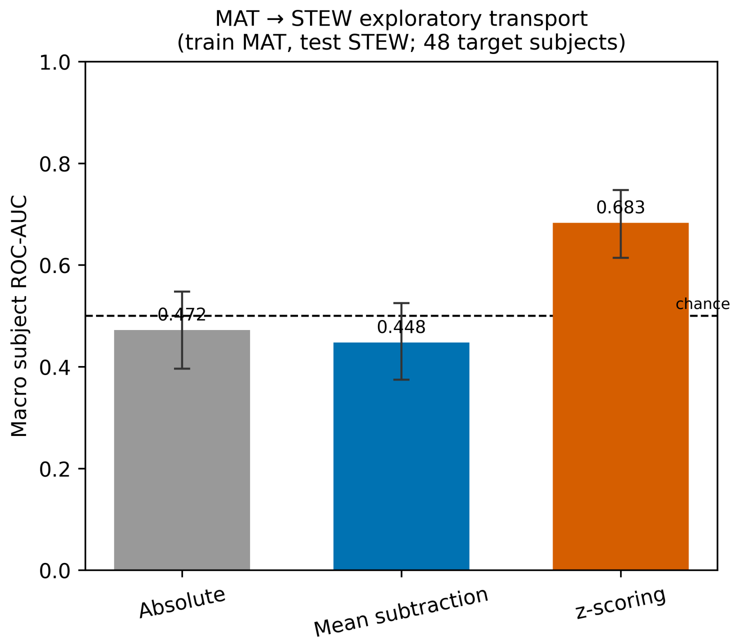

Figure 3: MAT→STEW exploratory transport. The predeclared candidate (mean subtraction) is below chance; only the post-hoc z-scoring diagnostic exceeds chance. Error bars are subject-bootstrap 95% CIs.

References

[1] Klimesch W. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. Brain Research Reviews. 1999;29(2-3):169-195.

[2] Gevins A, Smith ME, McEvoy L, Yu D. High-resolution EEG mapping of cortical activation related to working memory: effects of task difficulty, type of processing, and practice. Cerebral Cortex. 1997;7(4):374-385.

[3] Sridhar S, Manian V, Rao RPN. Theta and alpha ratios as objective correlates of cognitive workload: a cross-task replication study. Journal of Neural Engineering. 2022;19(4):046033.

[4] Zyma I, Tukaev S, Seleznov I, Kiyono K, Popov A, Chernykh M, Shpenkov O. Electroencephalograms during Mental Arithmetic Task Performance. Data. 2019;4(1):14.

[5] Lotte F, Bougrain L, Cichocki A, Clerc M, Congedo M, Rakotomamonjy A, Yger F. A review of classification algorithms for EEG-based brain-computer interfaces: a 10 year update. Journal of Neural Engineering. 2018;15(3):031005.

[6] Zyma I, Tukaev S, Seleznov I, Kiyono K, Popov A, Chernykh M, Shpenkov O. EEG During Mental Arithmetic Tasks [Dataset]. PhysioNet. 2019. https://doi.org/10.13026/C2JQ1P.

[7] Lim WL, Sourina O, Wang LP. STEW: Simultaneous task EEG workload data set. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2018;26(11):2106-2114.

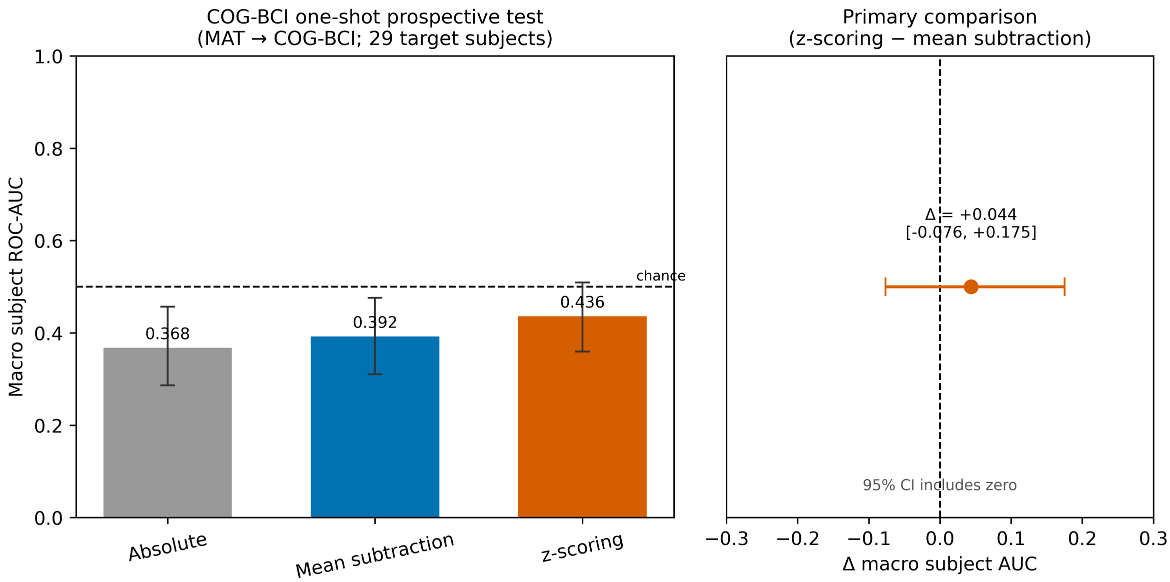

Figure 4: COG-BCI one-shot prospective test. Left: macro subject-level ROC-AUC by calibration mode (all below chance). Right: the primary paired contrast (z-scoring − mean subtraction) with 95% CI including zero. Error bars are subject-bootstrap 95% CIs; dashed lines mark chance (left) and zero (right).

[8] Hinss MF, Somon B, Jahanpour ES, Lafont du Sordet T, Verdiere KJ, Lartigue A, et al. Open multi-session and multi-task EEG cognitive dataset for passive brain-computer interface and mental workload applications (COG-BCI). Scientific Data. 2023;10:85. Dataset: Zenodo. https://doi.org/10.5281/zenodo.6874128.

[9] Barras M, Booth L. Cognitive Workload 8-level Arithmetic Task. OpenNeuro DS007262. 2026. https://doi.org/10.18112/openneuro.ds007262.v1.0.6.

[10] Gramfort A, Luessi M, Larson E, Engemann DA, Strohmeier D, Brodbeck C, Goj R, Jas M, Brooks T, Parkkonen L, Hamalainen M. MEG and EEG data analysis with MNE-Python. Frontiers in Neuroscience. 2013;7:267.

[11] Jayaram V, Alamgir M, Altun Y, Scholkopf B, Grosse-Wentrup M. Transfer learning in brain-computer interfaces. IEEE Computational Intelligence Magazine. 2016;11(1):20-31.

[12] Hakimi N, Jodeiri A, Setarehdan SK. Mental arithmetic task classification using convolutional neural networks on EEG signals. Biomedical Signal Processing and Control. 2023;79:104156.

[13] Roy RN, Charbonnier S, Campagne A, Bonnet S. Efficient mental workload estimation using task-independent EEG features. Journal of Neural Engineering. 2016;13(2):026019.

[14] Jebelli H, Hwang S, Lee S. EEG-based workers' mental workload classification using deep learning. Automation in Construction. 2019;106:102876.

[15] Nguyen PK, Huynh QL. A machine learning approach in EEG-based assessment of cognitive load by mental arithmetic tasks. Journal of Physics: Conference Series. 2025;2949:012004.

[16] Borghini G, Astolfi L, Vecchiato G, Mattia D, Babiloni F. Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness. Neuroscience & Biobehavioral Reviews. 2014;44:58-75.

[17] Cavanagh JF, Shackman AJ. Frontal midline theta reflects anxiety and cognitive control: a meta-analytic review. Journal of Physiology-Paris. 2014;108(4-6):251-258.

[18] Sweller J. Cognitive load during problem solving: effects on learning. Cognitive Science. 1988;12(2):257-285.

[19] Hart SG, Staveland LE. Development of NASA-TLX (Task Load Index): results of empirical and theoretical research. Advances in Psychology. 1988;52:139-183.

[20] Gramfort A, Luessi M, Larson E, Engemann DA, Strohmeier D, Brodbeck C, Goj R, Jas M, Brooks T, Parkkonen L, Hamalainen M. MEG and EEG data analysis with MNE-Python. Frontiers in Neuroscience. 2013;7:267.

[21] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research. 2011;12:2825-2830.

[22] Zander TO, Kothe C. Towards passive brain-computer interfaces: applying brain-computer interface technology to human-machine systems in general. Journal of Neural Engineering. 2011;8(2):025005.

[23] Vidaurre C, Sannelli C, Muller KR, Blankertz B. Co-adaptive calibration to improve BCI performance. NeuroImage. 2011;55(4):1443-1456.

[24] Congedo M, Barachant A, Bhatia A. Riemannian geometry for EEG-based brain-computer interfaces: a primer and a review. Brain-Computer Interfaces. 2017;4(3):155-174.

[25] Baldwin CL, Penaranda BN. Adaptive training using an artificial neural network and EEG metrics for within- and cross-task workload classification. NeuroImage. 2013;59(1):48-56.

[26] Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, et al. PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals. Circulation. 2000;101(23):e215-e220.

[27] Wong B. Points of view: Color blindness. Nature Methods. 2011;8(6):441.

[28] Zanini P, Congedo M, Jutten C, Said S, Berthoumieu Y. Transfer learning: a Riemannian geometry framework with applications to brain-computer interfaces. IEEE Transactions on Biomedical Engineering. 2018;65(5):1107-1116.